

Tanmay Saxena

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Summary

Backend Engineer with 3+ years of experience building high-throughput distributed systems, real-time WebSocket protocols, and scalable data pipelines. Proficient in Go, Python, Node.js, TypeScript with deep PostgreSQL optimization expertise and cloud infrastructure (AWS, GCP, Firebase).

Technical Skills

Languages: Go, Python 3.11+, TypeScript, JavaScript, Java, SQL
Backend: Node.js, Express.js, FastAPI (async), Django, WebSocket protocol design, REST/GraphQL APIs
Databases: PostgreSQL (pgvector, partitioning, indexing), MongoDB, MySQL, Redis, Prisma ORM, SQLAlchemy 2.0
Cloud & DevOps: AWS (EC2, Lambda, S3, RDS, CloudFormation), GCP, Firebase (Firestore, Cloud Functions), Docker
Architecture: Microservices, event-driven systems, distributed rate limiting, CI/CD (GitHub Actions), monorepo
Data & ML Infra: Alembic migrations, pgvector cosine similarity, Modal (serverless GPU), ONNX Runtime, Upstash Redis

Experience

Q IT Technologies

June 2025 – Present

Lead Software Developer (Backend)

Gainesville, FL

Directly — Real-Time AI Dance Coaching Platform

- Designed a typed bidirectional WebSocket protocol with 33+ discriminated message types covering real-time pose frames, staged feedback with per-joint deviations, voice commands, base64 TTS audio, rep tracking, and full session lifecycle control.
- Architected a privacy-first backend where no video leaves the client device — only 165 floats/frame transmitted over WebSocket; server processes pose data with <150ms end-to-end feedback latency.
- Set up PostgreSQL with async SQLAlchemy 2.0 + Alembic migrations + pgvector extension, storing coaching episodes with effectiveness scores, 1024-dim embedding vectors, and full MotionBlueprints as JSONB.
- Built a RAG pipeline backend — Voyage AI embedding task runs on 60s background cycle; cosine similarity retrieval ranked by weighted effectiveness scoring for personalized coaching.
- Architected a multi-stage teaching state machine (demo → practice → evaluate → full dance) with rep tracking, auto-advance at ≥70% score, idle detection, and real-time RNN beat detection via madmom with hierarchical choreography segmentation.

Shorten — AI Video Intelligence Platform

- Designed a 30+ model PostgreSQL schema supporting multi-tenant data isolation across YouTube, TikTok, and Instagram with audit trails, status tracking, and time-series engagement metrics.
- Implemented distributed rate limiting with Upstash Redis and feature-flag-based cost controls (max video duration, model selection) to keep per-brand analysis cost under \$5.
- Built Puppeteer-based stealth web scraping for TikTok as a zero-API-key fallback; developed 15+ CLI scripts for end-to-end pipeline orchestration (setup → ingestion → aggregation → report generation).
- Architected a 6-stage LLM pipeline generating enterprise reports from aggregated video signals, with graceful fallback chain (Gemini → Deepgram+Ollama+Groq → GPT-4o).

FavorIt — Peer-to-Peer Marketplace App

- Architected real-time communication platform using Node.js and Firebase (60% latency reduction); built Cloud Functions for payment processing, push notifications, and OpenAI-powered receipt OCR.
- Integrated Stripe SDK (PaymentIntent, manual capture, KYC via Stripe Identity); implemented Firestore real-time subscriptions with composite indexes and multi-document atomic transactions.

Applyin.co

Jun 2024 – Aug 2024

SDE Intern

Remote

- Built and optimized RESTful APIs in Node.js/Express.js, improving latency by 35%; implemented JWT auth flows and Google SSO integration, reducing onboarding time by 60%.

Tata Consultancy Services (TCS)

Jan 2021 – May 2023

Systems Engineer (Developer – General Motors)

Bengaluru, India

- Drove database architecture for a 1TB+ PostgreSQL dataset (indexing + partitioning, 60% query performance gain); automated data processing pipeline with Django, reducing manual work by 87.5%.
- Implemented CI/CD pipelines with GitHub Actions, cutting deployment time by 50%.

Education

University of Florida

Aug 2023 – May 2025

M.S. in Computer Science (GPA: 3.90/4.00)

Gainesville, FL

SRM Institute of Science and Technology

Jun 2016 – May 2020

B.Tech in Electronics and Communications Engineering (GPA: 8.58/10.00)

Chennai, India